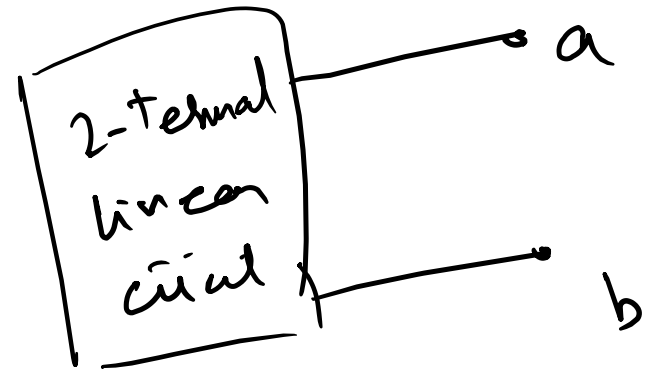
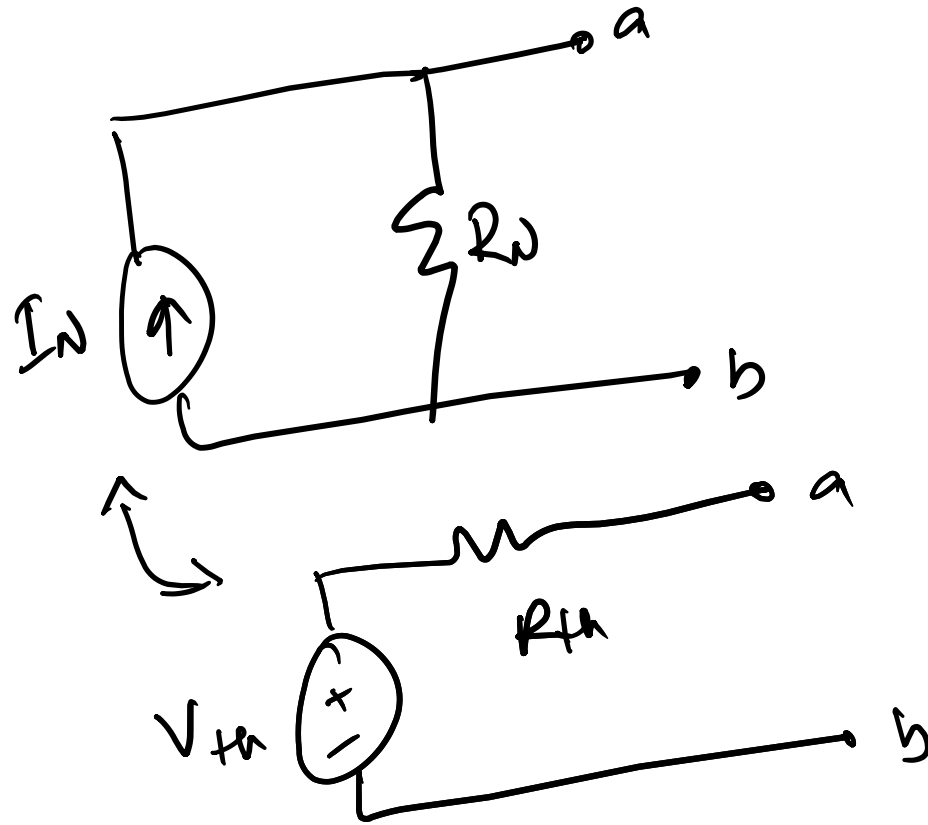


Norton's theorem:

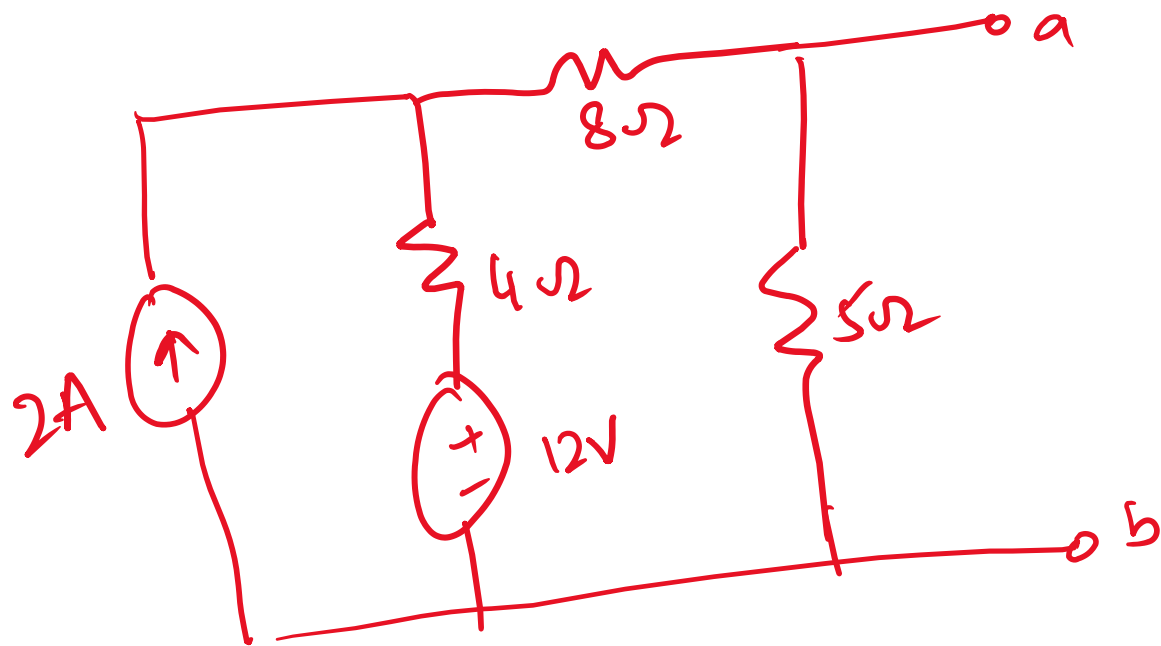


$$R_N = R_{th}$$

$$V_{th} = I_N \cdot R_{th}$$

I_N : Norton current or I_{sc} (Short circuit current)

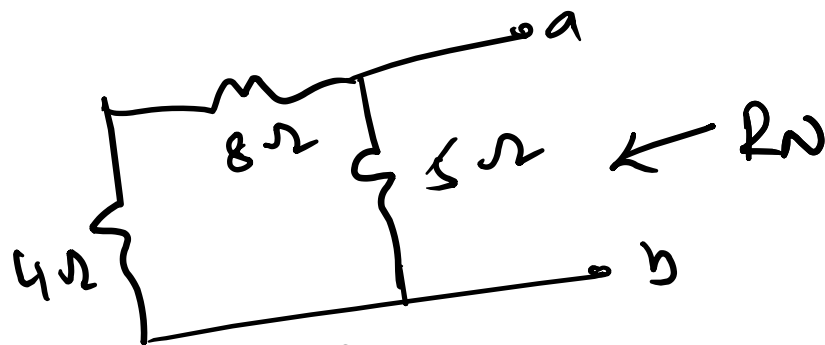
- * Remove load
- * Short the terminals, find the current flowing through them.



Find the Norton equivalent at a-b.

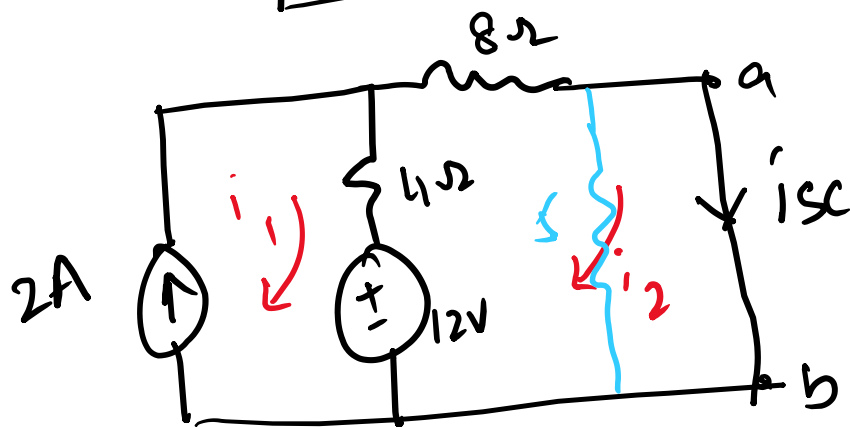
R_N , I_N (or I_{sc})

R_N :



$$R_N = (8 + 4) \parallel (5)$$

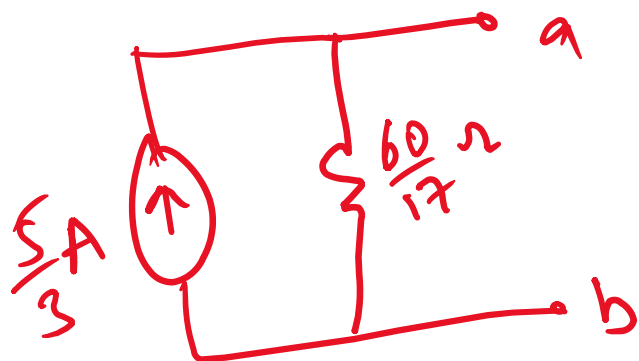
I_N :



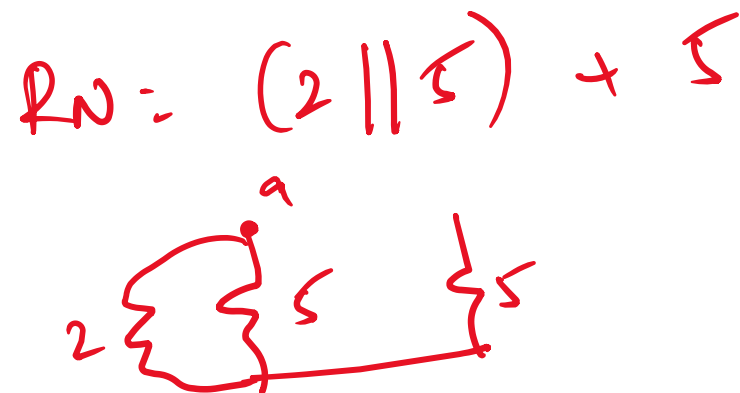
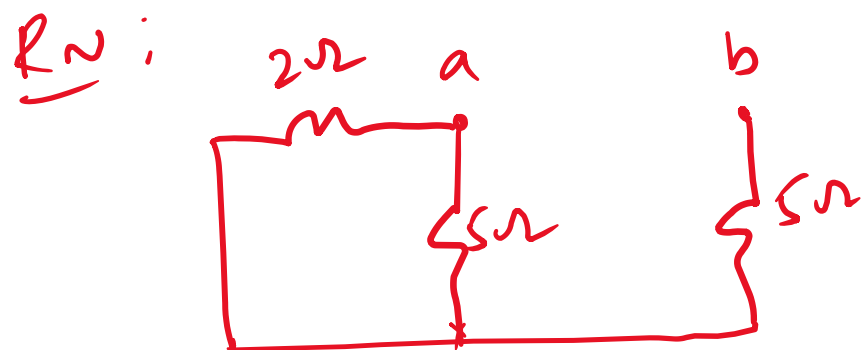
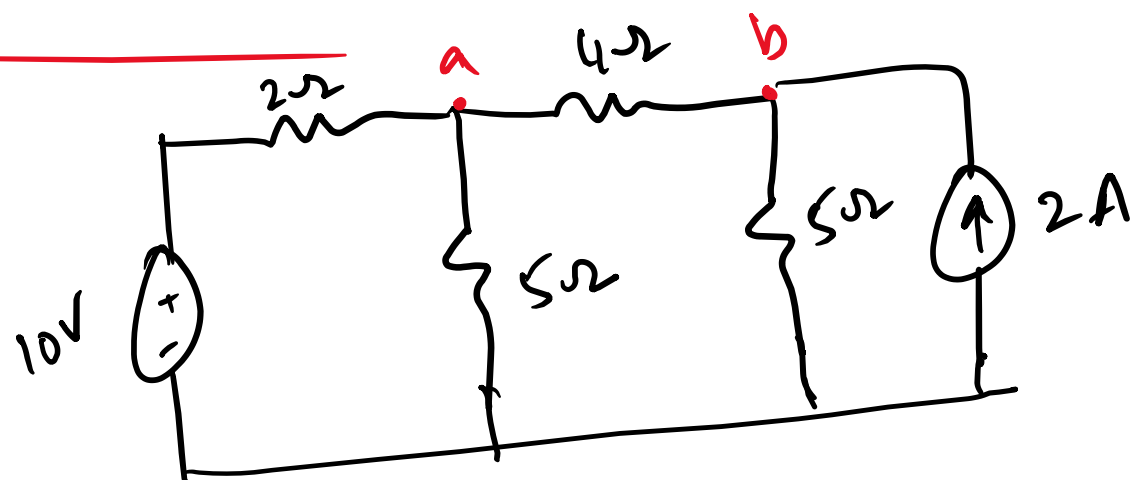
$$i_1 = 2A \quad ; \quad i_{sc} = i_2$$

$$12 - 4(i_2 - i_1) - 8i_2 = 0$$

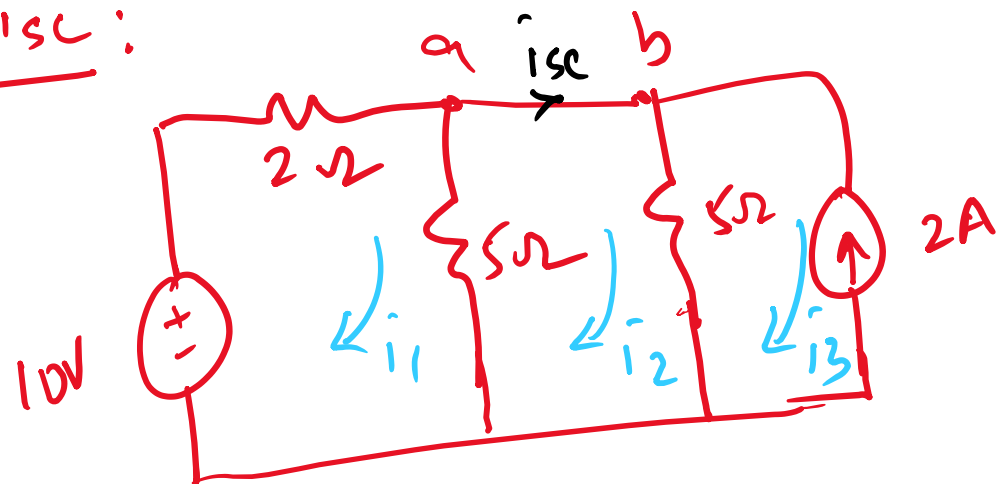
$$i_2 = \frac{5}{3}A = I_N$$



Find the Norton equivalent at 4Ω resistor.



$I_N = i_{sc}:$



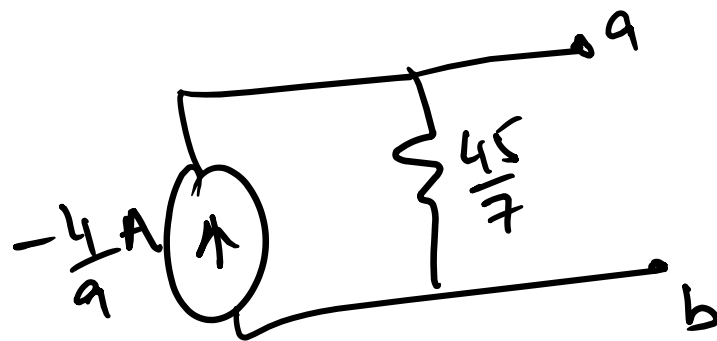
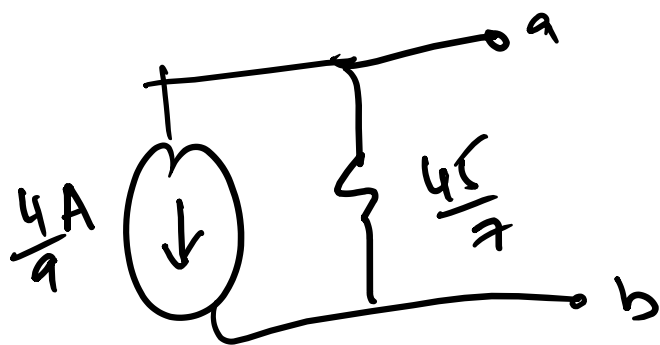
$$\hat{i}_3 = -2A$$

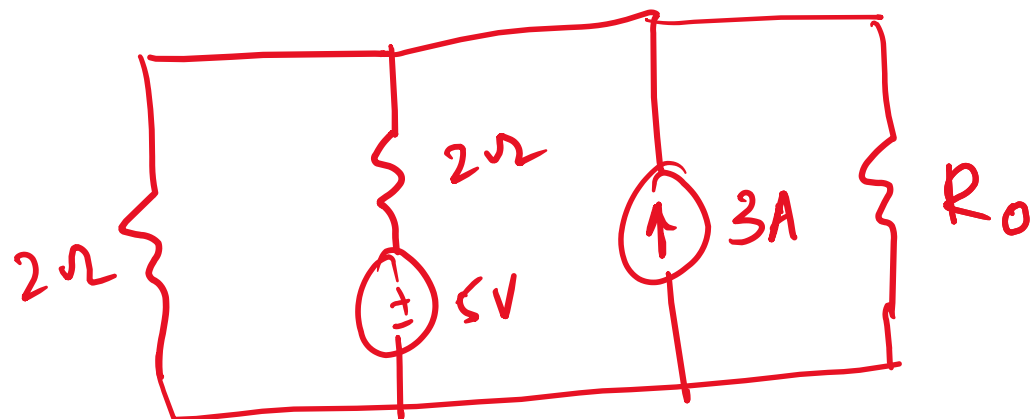
$$\hat{i}_{sc} = \hat{i}_2$$

$$10 - 2\hat{i}_1 - 5(\hat{i}_1 - \hat{i}_2) = 0$$

$$-5(\hat{i}_2 - \hat{i}_1) - 5(\hat{i}_2 - \hat{i}_3) = 0$$

$$\hat{i}_2 = -\frac{4}{9} = \hat{i}_{sc} = \hat{i}_N$$

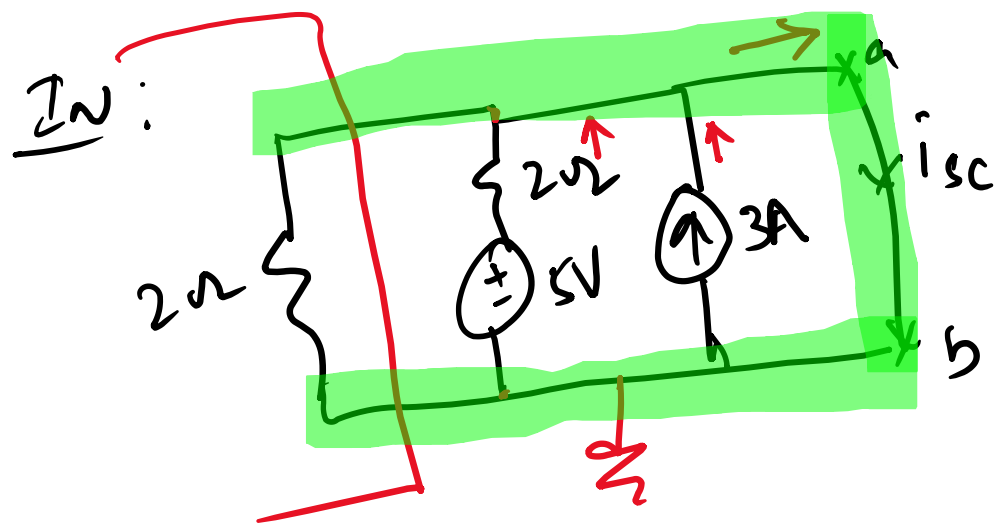




Find Norton equivalent at R_o



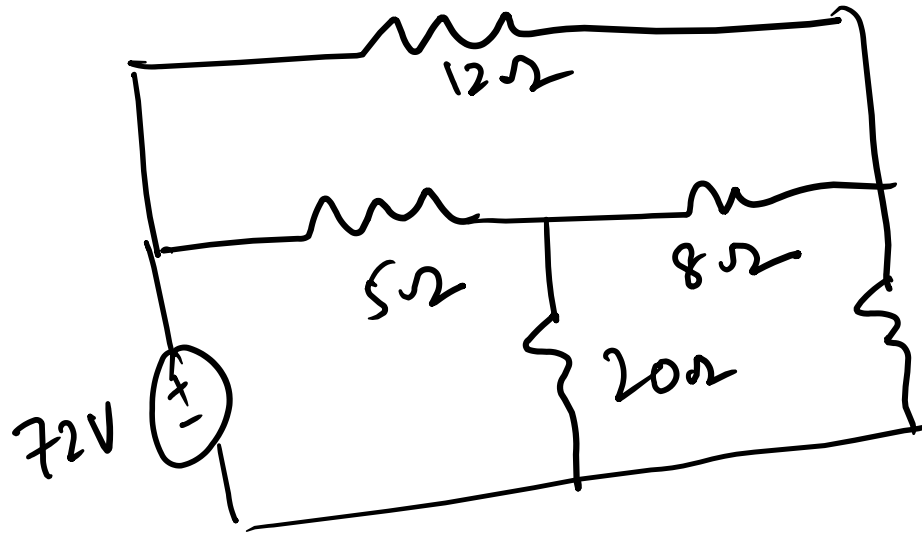
$$R_N = 1\Omega$$



$$V_a = V_b$$

$$i_{2\Omega} = 0$$

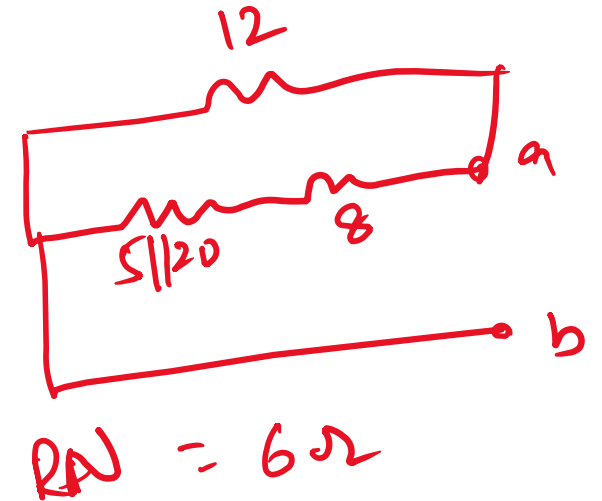
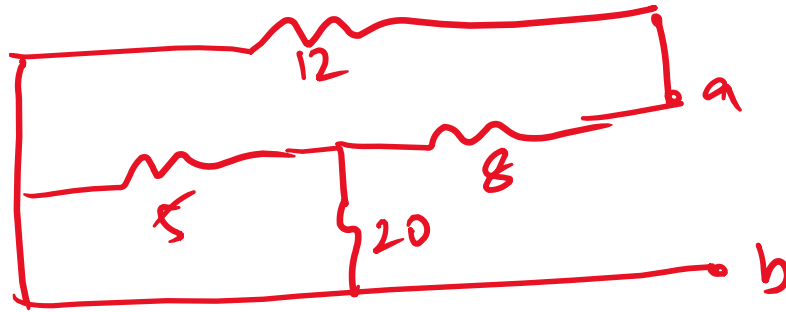
$$3 + \frac{5-0}{2} = i_{sc}$$



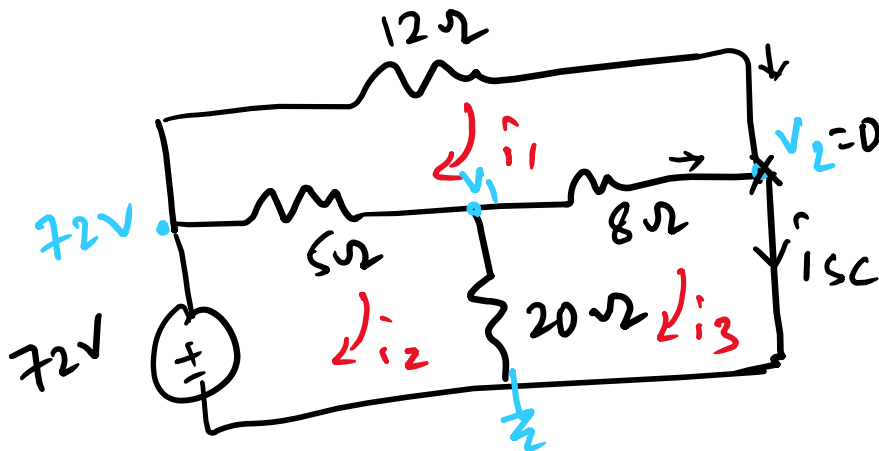
Find Norton equivalent
at 4Ω resistor.
Then calculate V_o .

4Ω V_o

R_N :



I_N :

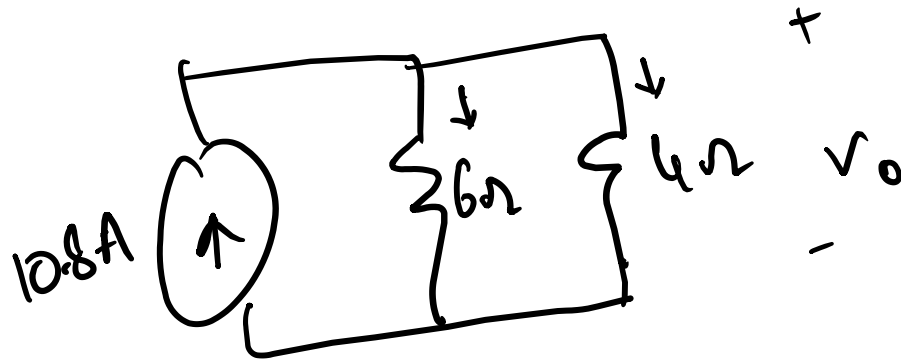


$$i_{sc} = i_3$$

$$\frac{72 - V_1}{5} = \frac{V_1}{20} + \frac{V_1}{8}$$

$$\frac{72}{12} + \frac{V_1}{8} = \hat{i}_{sc}$$

$$\hat{i}_{sc} = 10.8 \text{ A} = \hat{I}_N$$



$$10.8 = \frac{V_0}{6} + \frac{V_0}{4}$$

$$V_0 = \underline{\underline{25.9 \text{ V}}}$$